Will 20cm SiC WAFER Be The Next Standard?

Silicon carbide (SiC) has a lot of potential to replace silicon in many applications, a significant one being in the automotive industry, especially for battery-operated electric vehicles. To reduce the cost of producing SiC devices, which is quite high, SiC fabs are moving from 15cm (6-inch) to 20cm (8-inch) wafers

Porsche Carrera GT uses carbon-fibre-reinforced silicon carbide (C/SiC) ceramic composite braking system

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The semiconductor silicon carbide (SiC), commonly known as carborundum, occurs in nature as the mineral moissanite. It is actually abundantly found in space in the form of stardust but occurs very rarely in nature. Nonetheless, since 1893, SiC is being mass-produced for use as an abrasive or in manufacture of light emitting diodes (LEDs) and radio detectors. Today, the ceramics made with SiC, being durable and having high thermal shock resistance, are widely used in automotive brakes and clutches.

Why have we not been using SiC instead of silicon for making semiconductor devices? Earlier, the main problem with SiC crystals was that they had some defects that prevented them from being used widely in electrical applications. Hence, silicon became the standard. However, with technological advancements in production processes, 15cm SiC wafers are now giving a good yield. Particularly in the automotive industry, using SiC instead